

Recurrence Relation

A recurrence relation is an equation/formula which recursively defines any sequence i.e. in which any term of the sequence is function of its previous terms

ex $s_n = \{2^n\}, n \geq 0$

1, 2, 2^2 , 2^3 , ...

$$a_n = 2a_{n-1}, n=1, a_0=1$$

ex = 0, 1, 2, 3, 5, 8, 13, 21, 34, ...

$$a_n = a_{n-1} + a_{n-2}, a_0=0, a_1=1$$

• order of recurrence relation

eg: $a_n + 3a_{n-1} + 4a_{n-2} + 4a_{n-3} = f(n)$

$$n - (n-3) = \boxed{3} \text{ order}, \boxed{D=1}$$

The order of any recurrence relation is the difference of highest and lowest subscript of a .

Degree is the power of the highest subscript term.

Linear recurrence relation with constant coefficient

all terms are having power 1

and no two terms are in multiplication

e.g. $a_{n+1} = a_n \times a_{n-1}$ ~~X~~

$a_n^2 + a_{n-1} + a_{n-2} = f(n)$ ~~X~~

$a_{n+1} + b_1 a_{n+2} + b_2 a_{n+3} = f(n)$ ✓

Linear recurrence relation with constant coefficient of order n

$b_0 a_n + b_1 a_{n-1} + b_2 a_{n-2} + \dots + b_n a_0 = f(n)$

$b_0, b_1, b_2, \dots$ are constants

Homogeneous

$f(n) = 0$

Non-Homogeneous

$f(n) \neq 0$

solution of homogeneous recurrence relation

let $b_0 a_n + b_1 a_{n-1} + b_2 a_{n-2} + \dots + b_n a_0 = 0$

be the given recurrence relation then

to relation to find the solution

replace $a_n \rightarrow m^n$, $a_{n-1} \rightarrow m^{n-1}$ and so on
 to form auxiliary equation in terms of m
 we find the roots of auxiliary equation

Case 1: All the roots are distinct

i.e. $m_1, m_2, m_3, \dots$

$$a_n = C_1 \cdot m_1^n + C_2 m_2^n + C_3 m_3^n + \dots$$

Case 2: Some or all the roots are repeated i.e. $m_1, m_1, m_2, \dots$

$$a_n = (C_1 + n C_2) m_1^n + C_3 \cdot m_2^n$$

$$a_n = (C_1 + n C_2 + n^2 C_3) m_1^n + C_4 \cdot m_2^n$$

Case 3: If the roots are imaginary

$$a_n = C_1 (\alpha + i\beta)^n + C_2 (\alpha - i\beta)^n$$

repeated same as real roots

Methods

1) Iteration method X

2) Characteristic roots method ✓

3) Generating function method ✓

① eqn
② eqn

$$\text{ex } a_n - 3a_{n-1} + 2a_{n-2} = 0$$

$$\text{order} = 2 \quad [n - (n-2)] = 2$$

Af

$$m^2 - 3m + 2 = 0$$

$$m^2 - m - 2m + 2 = 0$$

$$m(m-1) - 2(m-1) = 0$$

$$(m-1)(m-2)$$

$$\boxed{m = 1, 2}$$

$$\boxed{a_n = C_1 + C_2 \cdot 2^n}$$

(2)

$$a_n = 3a_{n-1} + 4a_{n-2}, \quad a_0 = 0, a_1 = 7$$

$$a_n - 3a_{n-1} - 4a_{n-2} = 0$$

$$0 = -2$$

$$m^2 - 3m - 4 = 0$$

$$m^2 - 4m + m - 4$$

$$m(m-4) + 1(m-4) =$$

$$\boxed{m = 4, -1}$$

$$\boxed{a_n = C_1 \cdot 4^n + C_2 \cdot (-1)^n}$$

put $n=0$

$$a_0 \Rightarrow C_1 \cdot 4^0 + C_2 \cdot (-1)^0$$

$$0 \Rightarrow C_1 + C_2$$

put $n=1$

$$a_1 \Rightarrow C_1 \cdot 4^1 + C_2 \cdot (-1)^1$$

$$7 = 4C_1 - C_2$$

$$C_1 + C_2 = 0$$

$$4C_1 - C_2 = 5$$

$$5C_1 = 5$$

$$C_1 = 1$$

$$C_2 = -1$$

putting in eq

$$a_x = 4^x - (-1)^x$$

Solution of linear non-homogeneous recurrence relation with constant coeff

$$a_x = a_x(h) + a_x(p)$$

↓
left side
of recurrence relation

↓
right side
sol of Rnc relation

$$b_0 a_x + b_1 a_{x-1} + \dots + b_n a_0 = f(x)$$

Particular solution of Recurrence relation

Case I: if $f(x) = \text{const}$ then assume $a_x(p) = d_0 (\text{const})$

Case II: If $f(x) = C_0 + C_1 x$

Assume $a_x(p) = d_0 + d_1 x$

Case III: If $f(x) = C_0 + C_1 x + C_2 x^2$

Assume $a_x(p) = d_0 + d_1 x + d_2 x^2$

Case IV: $f(x) = C_0 m^x \rightarrow \textcircled{d_0} m^x$

$= (C_0 + C_1 x) m^x \rightarrow (d_0 + d_1 x) m^x$

$$(8) \quad a_x = 7a_{x-1} + 10a_{x-2} = 6 + 8x \quad \text{--- (A)}$$

Ans

$$m^2 - 7m + 10 = 0$$

$$m^2 - 5m - 2m + 10 = 0$$

$$m(m-5) - 2(m-5)$$

$$\boxed{m = 2, 5}$$

$$\boxed{a_x(h) = C_1 \cdot 2^x + C_2 \cdot 5^x}$$

Particular solution

$$\text{assume } a_x(p) = d_0 + d_1 x$$

$$a_x = d_0 + d_1 x$$

$$a_{x-1} = d_0 + d_1(x-1)$$

$$a_{x-2} = d_0 + d_2(x-2)$$

putting the value in (A)

$$(d_0 + d_1 x) - 7(d_0 + d_1(x-1)) + 10(d_0 + d_2(x-2)) = 6 + 8x$$

$$(d_0 - 7d_0 + 7d_1 + 10d_0 - 20d_1) \\ + (d_1 - 7d_1 + 10d_1)x = 6 + 8x$$

$$\Rightarrow (4d_0 - 13d_1) + 4d_1x = 6 + 8x$$

compare LHS & RHS

$$4d_0 - 13d_1 = 6$$

$$4d_1 = 8$$

$$\boxed{d_1 = 2}$$

$$\boxed{d_0 = 8}$$

putting in assumption

$$a_x(p) = 8 + 2x$$

$$\text{complete solution} = a_x(h) + a_x(p)$$

$$C.S = \boxed{C_1 \cdot 2^x + C_2 \cdot 5^x + 2x + 8}$$

$$(9) \quad a_x - 5a_{x-1} = 9$$

$$m + 5 = 0$$

$$\boxed{m = -5}$$

$$a_x(h) = C_1 \cdot 5^x$$

particular sol

$$\text{assume } a_x(p) = d_0 \text{ (const)}$$

$$a_x = d_0$$

$$a_{x-1} = d_0$$

putting value in (A)

$$d_0 + 5d_0 = 0$$

$$d_0 \Rightarrow \frac{0}{46}$$

$$d_0 = \frac{3}{2}$$

$$C.D = a_n(p) + a_n(r)$$

$$C.D = C_1(-5)^n + \frac{3}{2}$$

$$(3) \quad a_n - a_{n-1} = 6a_{n-2} = -30 \quad \text{--- (A)}$$

subjected to $a_0 = 20, a_1 = -5$

$$Ans = 11(-2)^n + 43^n + 5$$

= C.E

$$m^2 - m - 6 = 0$$

$$m^2 - 3m + 2m - 6 = 0$$

$$m(m-3) + 2(m-3) = 0$$

$$m = 3, -2$$

$$a_n(r) = C_1 \cdot 3^n + C_2 \cdot (-2)^n$$

$$a_0 \Rightarrow C_1 \cdot 3^0 + C_2 \cdot (-2)^0$$

$$20 = C_1 + C_2$$

$$a_1 = C_1 \cdot 3 + C_2 \cdot (-2)$$

$$-5 = 3C_1 - 2C_2$$

$$\begin{aligned} c_1 + c_2 &= 20 \\ 3c_1 - 2c_2 &= -5 \end{aligned}$$

$$\begin{aligned} 2c_1 + 2c_2 &= 40 \\ 3c_1 - 2c_2 &= -5 \end{aligned}$$

$$5c_1 = 35$$

$$\begin{aligned} c_1 &\Rightarrow 7 \\ c_2 &= 13 \end{aligned}$$

particula solu

$$a_n(p) = d_0$$

$$a_n = d_0$$

$$a_{n-1} = d_0$$

$$a_{n-2} = d_0$$

putting in (A)

$$d_0 + (-d_0) - 6(d_0) = -30$$

$$d_0 \Rightarrow 5$$

$$a_n = 5$$

$$c: s \Rightarrow a_n(p) + a_n(p)$$

$$= + 5$$

$$(4) \quad a_{x+2} - 4a_x = x^2 + x - 1 \quad \text{--- (A)}$$

$$m^2 - 4$$

$$A_2: \cancel{C_1 C_2 C_3 C_4} \text{ check}$$

$$\boxed{m = -2, 2}$$

$$a_x(h) = C_1(-2)^x + C_2(2)^x$$

Particular sol.

$$\text{assume } a_x(p) = d_0 + d_1 x + d_2 x^2$$

$$a_x \Rightarrow d_0 + d_1 x + d_2 x^2$$

$$a_{x+2} = d_0 + d_1(x+2) + d_2(x+2)^2$$

putting in (A)

$$\Rightarrow d_0 + d_1 x + d_2 x^2 - 4(d_0 + d_1(x+2) + d_2(x+2)^2) = x^2 + x - 1$$

$$d_0 + d_1 x + d_2 x^2 - 4(d_0 + d_1 x + d_1 \cdot 2 + d_2 x^2 + d_2 \cdot 4x + d_2 \cdot 4)$$

$$d_0 + d_1 x + d_2 x^2 - 4d_0 - 4d_1 x - 4d_1 \cdot 2 - 4d_2 x^2 - 4d_2 \cdot 4x - 4d_2 \cdot 4$$

$$= x^2 + x - 1$$

$$(d_0 - 4d_0 - 8d_1) + (d_1 - 4d_1 - 16d_2)x + (d_2 - 4d_2)x^2 = x^2 + x - 1$$

compare

$$-3d_0 - 8d_1 = 1$$

$$-3d_1 - 16d_2 = 1$$

$$d_2 - 4d_2 = -1$$

$$\boxed{d_2 = \frac{1}{3}}$$

$$\boxed{d_1 = -\frac{7}{9}}$$

$$\boxed{d_0 = -\frac{17}{27}}$$

putting in ans.

$$a_x(p) = -\frac{17}{27} + \frac{-7}{9}x - \frac{1}{3}x^2$$

$$c.s = a_x(h) + a_x(p)$$

$$\boxed{\Rightarrow C_1(-2)^n + C_2(2)^n - \frac{x^2}{3} - \frac{7}{9}x - \frac{17}{27}}$$

$$(0) \quad a_{x+2} - 5a_{x+1} + 6a_x = 5^x \quad \text{--- (A)}$$

Aux

$$m^2 - 5m + 6 = 0$$

$$m^2 - 2m - 3m + 6 = 0$$

$$m(m-2) - 3(m-2)$$

$$m = 2, 3$$

$$a_x(R) = C_1 \cdot 2^x + C_2 \cdot 3^x$$

Assume that

$$a_x(p) = d_0 \cdot 5^x$$

$$a_{x+1} = d_0 \cdot 5^{x+1}$$

$$a_{x+2} = d_0 \cdot 5^{x+2}$$

putting in (A)

$$d_0 \cdot 5^{x+2} - 5d_0 \cdot 5^{x+1} + 6d_0 \cdot 5^x = 5^x$$

$$\Rightarrow 5^x [d_0 \cdot 5^2 - 5d_0 \cdot 5 + 6d_0] = 5^x \cdot 1$$

$$25d_0 - 25d_0 + 6d_0 = 1$$

$$\boxed{d_0 \Rightarrow \frac{1}{6}}$$

putting in assumption

$$a_x(p) = \frac{1}{6} 5^x$$

$$c.s.: a_x(R) + a_x(p)$$

$$\Rightarrow \boxed{C_1 \cdot 2^x + C_2 \cdot 3^x + \frac{1}{6} 5^x} \quad Q$$

If any root are same with particular solution then we multiply it with x in particular solution

$$(8) a_x - 3a_{x-1} - 4a_{x-2} = 4^x \quad \text{--- (A)}$$

Ans

$$m^2 - 3m - 4 = 0$$

$$m^2 - 4m + m - 4 = 0$$

$$m(m-4) + (m-4)$$

$$\boxed{m = 4, -1}$$

$$a_x(h) = C_1(-1)^x + C_2 \cdot 4^x$$

Assume particular solution

$$a_x(p) = (4^x \cdot x) d_0$$

$$a_{x-1}(p) \Rightarrow d_0(x-1) 4^{x-1}$$

$$a_{x-2}(p) = d_0(x-2) 4^{x-2}$$

putting in (A)

$$d_0 \cdot x \cdot 4^x - 3(d_0(x-1) 4^{x-1}) - 4(d_0(x-2) 4^{x-2}) = 4^x$$

$$4^x \{ d_0 \cdot x - 3(d_0(4^{x-1} \cdot x - 4^{x-1})) - 4(d_0(x \cdot 4^{x-2} - 2 \cdot 4^{x-2})) \}$$

$$\Rightarrow 4^x \cdot d_0 x - 3 d_0 4^{x-1} \cdot x + 3 d_0 4^{x-1} - 4 d_0 x \cdot 4^{x-2} + 8 \cdot 4^{x-2}$$

$$\Rightarrow 4^x [d_0 x - 3 d_0 \cdot 4^{-1} \cdot x + 3 d_0 \cdot 4^{-1} - d_0 x \cdot 4^{-1} + 8 \cdot 4^{-2}]$$

$$= 4^x [d_0 x - 4 d_0 4^{-1} \cdot x + 3 d_0 4^{-1} + 1] \cdot \frac{1}{2}$$

$$3 \text{ do } 4^{-1} + \frac{1}{2} \Rightarrow 1$$

$$\frac{3}{4} \text{ do } + \frac{1}{2} = 1$$

$$\frac{3}{4} \text{ do } 1 - \frac{1}{2}$$

$$\text{do} \Rightarrow \frac{1}{2} \times \frac{4}{3}$$

$$\boxed{\text{do} = \frac{2}{3}}$$

$$(9) \quad a_n - a_{n-1} - 12a_{n-2} = (-3)^n + 6 \cdot 4^n$$

Ans

$$m^2 - m - 12 = 0$$

$$m^2 - 4m + 3m - 12 = 0$$

$$m(m-4) + 3(m-4)$$

$$\boxed{m = -3, 4}$$

$$a_n(p) = c_1(-3)^n + c_2 4^n$$

Assume auxi

$$a_n(p) = d_0 n (-3)^n + d_1 4^n \cdot n$$

$$a_{n-1}(p) = d_0 (n-1) (-3)^{n-1} + d_1 (n-1) 4^{n-1}$$

$$a_{n-2}(p) = d_0 (n-2) (-3)^{n-2} + d_1 (n-2) 4^{n-2}$$

putting in (A)

$$\left[d_0 n (-3)^n + d_1 4^n \cdot n \right] - \left[d_0 (n-1) (-3)^{n-1} + d_1 (n-1) 4^{n-1} \right] \\ - 12 \left[d_0 (n-2) (-3)^{n-2} + d_1 (n-2) 4^{n-2} \right]$$

$$\Rightarrow d_0 n (-3)^n - d_0 (n-1) (-3)^{n-1} - 12 d_0 (n-2) (-3)^{n-2} \\ + d_1 n \cdot 4^n - d_1 (n-1) 4^{n-1} - 12 d_1 (n-2) 4^{n-2} \\ = (-3)^n + 6 \cdot 4^n$$

$$\Rightarrow d_0 (-3)^n \left[n - \frac{n-1}{-3} - 12 \frac{(n-2)}{9} \right] \cdot$$

$$+ d_1 4^n \left[n - \frac{(n-1)}{4} - 12 \frac{(n-2)}{16} \right] = (-3)^n + 6 \cdot 4^n$$

compare LHS = RHS

$$\Rightarrow d_0 \left(n + \frac{(n-1)}{3} - 12 \frac{(n-2)}{9} \right) = 1$$

$$d_0 \left(\frac{9n + 3(n-1) - 12(n-2)}{9} \right) = 1$$

$$\frac{21 d_0}{9} = 1 \quad \boxed{d_0 = \frac{3}{7}}$$

$$d_1 \left(n - \frac{(n-1)}{4} - 12 \frac{(n-2)}{16} \right) = 6$$

$$d_1 \left(\frac{16n - 4n + 4 - 12(n-2)}{16} \right) = \frac{28 d_1}{16} = 6$$

$$\boxed{d_1 = \frac{24}{7}}$$

$$C.S \Rightarrow a_n(h) + a_n(p)$$

$$C.S \Rightarrow c_1(-3)^n + c_2(4)^n + \frac{3}{7}n(-3)^n + \frac{24}{7}n(4)^n$$

$$(9) \quad a_n - 4a_{n-1} + 4a_{n-2} = (n+1)2^n$$

Aux

$$m^2 - 4m + 4 = 0$$

$$m^2 - 2m - 2m + 4 = 0$$

$$m(m-2) - 2(m-2)$$

$$\boxed{m=2, 2}$$

$$a_n(h) \Rightarrow (c_1 + nc_2)(2)^n$$

Assuming particular, s

$$a_n(p) \Rightarrow d_0 + d_1 n$$

$$\Rightarrow (d_0 + d_1 n) \cdot 2^n$$

$$= (d_0 + d_1 n) \cdot n \cdot 2^n$$

$$a_{n-1} \Rightarrow (d_0 + d_1(n-1)) (n-1) 2^{n-1}$$

$$a_{n-2} = (d_0 + d_1(n-2)) (n-2) 2^{n-2}$$

putting in (A)

$$(d_0 + d_1 x) x^2 \cdot 2^x - 4 [d_0 (x-1)^2 + d_1 (x-1)^3] 2^{x-1} + 4 [d_0 (x-2)^2 + d_1 (x-2)^3] 2^{x-2} = (1-x) 2^x$$

$$\Rightarrow (d_0 x^2 + d_1 x^3) 2^x - [4 d_0 (x^2 + 1 - 2x) - 4 d_1 (x^3 - 1 - 3x^2 + 3x)] 2^{x-1}$$

$$+ [4 d_0 (x^2 + 4 - 4x) + 4 d_1 (x^3 - 8 - 6x^2 + 12x)]$$

$$2^{x-2} = (1+x) \cdot 2^x$$

$$\Rightarrow 2^x [d_0 x^2 + d_1 x^3 + \frac{(-4 d_0 x^2 - 4 d_0 + 8 d_0 x) + (-4 d_1 x^3 + 4 d_1 + 12 d_1 x^2 - 12 d_1 x)}{2}]$$

$$+ \frac{(4 d_0 x^2 + 16 - 16 x) + (4 d_1 x^3 - 32 d_1 - 24 d_1 x^2 + 48 d_1 x)}{4}$$

$$\Rightarrow (1+x) 2^x$$

$$\Rightarrow 2^x [\cancel{d_0 x^2} + \cancel{d_1 x^3} - 8 \cancel{d_0 x^2} - 8 \cancel{d_0} + 16 \cancel{d_0 x} - 8 \cancel{d_1 x^3} + 8 \cancel{d_1} + 24 \cancel{d_1 x^2} - 24 \cancel{d_1 x} + 4 \cancel{d_0 x^2} + 16 \cancel{d_0} - 16 \cancel{d_0 x} + 4 \cancel{d_1 x^3} - 32 \cancel{d_1} - 24 \cancel{d_1 x^2} + 48 \cancel{d_1 x}]$$

$$= 2^x \left[\begin{array}{l} -8 d_0 + 16 d_0 \\ -24 d_1 x + 48 d_1 x \end{array} + 16 d_0 x + \cancel{16 d_0 x} + 8 d_1 - 32 d_1 - 16 d_1 x \right]$$

$$2^x [8 d_0 - 24 d_1 - 24 d_1 x]$$

NOTHING

$$\frac{(2d_0 - 24d_1 + 24d_1x)}{4} z^r = (1+x)/2^r$$

$$(2d_0 - 6d_1 + 6d_1x) = (1+x)$$

$$2d_0 - 6d_1 = 1$$

$$d_1 = \frac{1}{6}$$

$$2d_0 - 8 \times \frac{1}{6} = 1$$

$$2d_0 = 2$$

$$d_0 = 1$$

$$a_x(p) = \left(1 + \frac{x}{6}\right) x^2 2^r$$

$$c.s = a_x(h) + a_x(p)$$

$$c.s = \left[(1+c_2x) z^r + \left(1+\frac{x}{6}\right) x^2 2^r \right]$$

$$p) a_n - 4a_{n-1} + 3a_{n-2} = n^2$$

$$m^2 - 4m + 3 = 0$$

$$m = 3, 1$$

$$a_x(h) = c_1(3)^n + c_2(1)^n$$

Assume p.s

$$a_x(p) = d_0 + d_1 n + d_2 n^2$$

NOTHING

$$a_n(p) = d_0 n + d_1 n^2 + d_2 n^3$$

$$a_{n-1}(p) = d_0(n-1) + d_1(n-1)^2 + d_2(n-1)^3$$

$$a_{n-2}(p) = d_0(n-2) + d_1(n-2)^2 + d_2(n-2)^3$$

putting (A)

$$d_0 n + d_1 n^2 + d_2 n^3 - 4 [d_0(n-1) + d_1(n-1)^2 + d_2(n-1)^3] + 3 [d_0(n-2) + d_1(n-2)^2 + d_2(n-2)^3] = n^2$$

$$\Rightarrow d_0 n + d_1 n^2 + d_2 n^3 + [-4 d_0 n^2 + 4 d_0 - 4 d_1 n^2 - 4 d_1 + 8 d_1 n - 4 d_2(n^3 - 1 - 3n^2 + 3n)] + [3 d_0 n - 6 d_0 + 3 d_1 n^2 + 12 d_1 - 12 d_1 n + 3 d_2(n^3 - 8 - 6n^2 + 12n)] = n^2$$

$$\Rightarrow \cancel{d_0 n} + d_1 n^2 + \cancel{d_2 n^3} - 4 d_0 n^2 + 4 d_0 - 4 d_1 n^2 - 4 d_1 + 8 d_1 n - 4 d_2 n^3 + 4 d_2 + 12 d_2 n^2 - 12 d_2 n + 3 d_0 n - 6 d_0 + 3 d_1 n^2 + 12 d_1 - 12 d_1 n + 3 d_2 n^3 - 24 d_2 - 18 d_2 n^2 + 36 d_2 n = n^2$$

$$\Rightarrow \cancel{d_0 n} + 3 d_0 n + \cancel{d_1 n^2} \Rightarrow -6 n^2 d_2 - 48 n d_2 + 20 d_2$$

$$\frac{-6 d_2 n^2}{n^2} = n^2 \quad \text{compare}$$

$$d_2 = -\frac{1}{6}$$